

ration with DIY Workshop, Hyderabad. Virtual Braille is a device that converts digital text to Braille in real-time allowing the visually challenged to have tactile Braille feedback to the tracked finger thereby discerning unique letters and words. Needless to say, this low-cost eBook reader for the visually challenged has truly disruptive capabilities, especially when existing Braille printing costs several lakhs of rupees and text-to-speech softwares aren't accurate or localized.

Angad calls himself a maker, as he's been obsessed with making things from a very young age – taking apart toy cars, building structures out of lego blocks, doing a lot of art work, designing automobiles, soldering, etc. His goal in life? “To try to start the Maker Movement (<http://www.techopedia.com/definition/28408/maker-movement>) in India to give rise to youngsters who are more passionate about and who wish to solve real world problems through technology.” In a suburb in Mumbai, along with the help of like-minded enthusiasts, he's even helped start the Maker's Asylum, a place where people can come and just make things. It's fully equipped like a professional workshop, the intention of which is to give anyone who wants to transform their idea into a tangible item a fair shot at making it happen.

He also has a website (www.angad-makes.com) on which he uploads all the projects he's working on along with tutorials and open source instructions to help anyone interested to learn how his devices function, and even re-make them. Angad also conducts talks and events at schools to encourage kids to explore hands on learning through workshops – e.g. something as simple as water rocketry. He

FOOD FOR THOUGHT

What are some of the applications of 3D printing that you can think of? Can it be used to 3D print your own medicines? Can you, perhaps, make synthetic organs to help someone's life? Or print out prosthetics to enrich the lives of millions of people with disabilities? Can we 3D print our own garments, for instance, with synthetic fabric? Can we print out a basic circuit board or a transistor on a 3D printer? 3D printing is nothing short of a second Industrial Revolution, empowering not only industries and companies, but every single individual to take control of their lives like never before. We leave you to imagine the endless possibilities of this truly revolutionary technology.

has a startup which provides low-cost DIY kits to get young minds started off with DIY learning.

Being homeschooled from a young age, Angad has strong views on our education system as well. He says, “The problem is partially with the system and partially with people's mindset. The formal classroom education system does not allow a person to learn at one's own pace and innovate. Emphasis on exam performance corrupts the very purpose of education. Also, many people (parents/institutions) are not willing to invest time and money in youngsters who have curious minds.”

Angad respects technology, knowing fully well how it has inspired him to pursue his dreams. “Technology helps me learn and do things even without having to enroll in any particular institution or programme,” says Angad. “It helps me see science happen in front of my eyes and gets me more and more interested in solving

real world problems. Technology helps me to create technology. I can also share information and instruct people on how they can create the same technology as I have been using the technology available to them.” Angad also acknowledges the Internet as an instrument of self-empowerment and learning, and he owes 90% of his tech knowledge to the Internet.

We asked this truly remarkable 15-year-old on his vision and how his breakthroughs will help millions of Indians. “I'm trying to make technology cheaper, more portable and faster. Having low cost prototyping devices, low cost health diagnostic tools, low cost and fast communication, faster and cheaper educative tools, etc will make not just India but this world simpler, faster, reliable and better!”

Karan Chaphekar

Calling himself the Maker-in-Chief at KCbots.com, Karan Chaphekar is cut largely from the same cloth as Angad Daryani. Karan has been tinkering with hardware and dabbling in robotics for as long as he can remember. After getting his degree in Electronics Engineering, Karan, now in his early twenties, dedicates most of his time focusing his efforts on 3D printing. Karan and his team has been active in the maker community for the past three years, where they're utilising their expertise in the 3D printing and robotronics domains and coming out with a revolutionarily simple and powerful 3D printer. He claims it to be uber simple – works “out of the box” – while retaining a powerful print performance. It is called, simply, the KUBE.

The KUBE allows anyone with a desktop computer to “grow” and build a real object from plastic, by just clicking



Angad's soon-to-release 3D printer is a cool device to spend time around, no?



This is a 3D-printed plastic bracelet, produced by Karan's invention



As you can see through these 3D-printed scale models, the force is strong in Karan